

Problem Statement

Customer segmentation for targeted marketing is one of the most vital applications of the clustering algorithm. Here, as a manager of the online store, you would want to group the customers into different clusters, so that you can make a customised marketing campaign for each of the group. You do not have any label in mind, such as good customer or bad customer. You want to just look at patterns in customer data and then try and find segments. This is where clustering techniques can help you with segmenting the customers. Clustering techniques use raw data to form clusters based on common factors among various data points. This is exactly what will also be done in segmentation, where various people or products will be grouped together on the basis of similarities and differences between them.

As a manager, you would have to decide what the important business criteria are on which you would want to segregate the customers. So, you would need a method or an algorithm that itself decides which customers to group together based on these criteria.

Mainly 3 types of segmentation are used for customer segmentation:

- Behavioural segmentation: Segmentation is based on the actual patterns displayed by the consumer
- Attitudinal segmentation: Segmentation is based on the beliefs or the intents of people, which may not translate into similar action
- Demographic segmentation: Segmentation is based on a person's profile and uses information such as age, gender, residence locality, income, etc.

Code-Based Questions

1. Data Exploration

- How many total customers are present in the dataset?
- How many features are used for clustering after preprocessing?
- Which feature has the highest variance among all features?

2. Cluster Formation

- How many clusters were finally created by the clustering algorithm?
- What is the size (number of customers) of each cluster?
- Which cluster contains the highest number of customers?

3. Cluster Characteristics

- What is the average income of customers in each cluster?
- Which cluster has the highest average spending score?
- Which cluster represents the lowest spending customers?

4. Model Evaluation

- What is the **inertia value** obtained from the K-Means clustering model?
- What is the **silhouette score** of the clustering result?
- Based on the elbow method, what is the optimal number of clusters?

5. Visualization-Based Questions

- In the cluster visualization plot, which cluster appears most densely populated?
- Which cluster represents customers with **high income but low spending score**?
- Which cluster represents **low income but high spending customers**?

6. Cluster Label Analysis

- Which cluster would most likely represent **premium customers**?
- Which cluster might correspond to **price-sensitive customers**?

7. Outlier Detection

- Are there any potential outliers visible in the dataset after clustering?
- Which data points appear farthest from cluster centroids?

8. Centroid Analysis

- What are the centroid values for each cluster?
- Which cluster centroid has the highest average income value?